UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa
Department of Computing & Technology

Investigate, Recommend or Escalate: A Rule-Constrained Reinforcement Learning Agent for IT Incident Triage under Ambiguous Evidence

By

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A proposal submitted to the Faculty of Engineering Design & Technology for the study leading to the Degree in partial fulfilment of the requirements for the award of **Bachelor of Science in Computer Science** of Uganda Christian University

SPECIALIZATION TRACK: *Artificial Intelligence and Robotics*

IAN RAYMOND OSOLO

[Enter Version Date]

Declaration

I, *OPOL CHARIS*, declare to the best of my knowledge that the work in this proposal is original and has not been submitted before to any university/institution. It is the result of my effort.

Signature: _____

Date: _____

Ethical Generative AI Usage Declaration

Students may use generative artificial intelligence (AI) ethically as a "study buddy" or assistant rather than a replacement for critical thinking. This includes citing AI use and fact-checking all outputs to maintain academic integrity.

For purposes of this assignment, the following shall guidelines shall apply.

i. Generative AI may be used for brainstorming, and generating ideas to improve your work. You may also use AI for literature search, and summarizing journal papers for quick analysis and assessing relevance of publications. However, once the relevant publications have been identified, a student is expected to fully review the selected publications, identifying problems addressed, methods used, gaps identified and recommendations made.

ii. Where AI has been used as guided in (i) above, full disclosure of how it has been used should be made in the appendix C, including how AI was used, the AI tools (including mode versions), and a compilation on major AI prompts used.

iii. NO GENERATIVE AI CONTENT IS ALLOWED IN THE FINAL SUBMISSION. The final submission SHOULD be your original work.

A. **Declaring the non-use of AI for content generation**

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Do not sign this statement while the proposal contains text drafted with an AI assistant (see Appendix C). Rewrite that text in your own words first, or ask your supervisor how to declare it.

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B. **Acknowledging the use of AI as a study buddy**

Please refer to Appendix C for the complete declaration of AI usage for generating ideas, brainstorming, literature search, and summarizing only to improve your work.

Approval

This proposal has been submitted for examination with approval of the following advisors/supervisor.

IAN RAYMOND OSOLO
Department of Computing & Technology
Faculty of Engineering Design & Technology
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Signature: _____

Date: _____

Abstract

When an IT incident is reported, the staff who receive it must decide what to do before they know its cause: investigate further, apply a likely fix, or pass the incident to someone better placed to resolve it. The first evidence is often incomplete, unreliable or consistent with more than one cause, yet most automated triage tools treat the decision as a single prediction from the incident report. They cannot ask for more evidence, cannot say that they are unsure, and do not check their suggestions against an organisation's operational rules.

This research proposes to design and evaluate a rule-constrained reinforcement learning agent that, for each incident, learns whether to request a diagnostic check, recommend a fix that the operational rules permit, or escalate the incident to a human expert. An independent rule-based safety mechanism checks every proposal, returns a blocked proposal to the agent once, and escalates the incident if the second proposal is also blocked. The agent will be trained and evaluated in a simulation built from incident scenarios elicited from the Directorate of University ICT Services at Uganda Christian University, in which missing, unreliable and overlapping evidence can each be controlled, and compared with six baseline policies on decision quality, investigation efficiency, escalation behaviour and constraint compliance. A systematic review of 34 studies, conducted for this research, found no published agent that combines these three decisions for IT service incidents under explicit rules.

Dedication

I dedicate this research proposal to my father, Rev. Canon Dr Samuel Opol.

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Chapter 1

Introduction

1.1 Background

Organisations depend on IT services that are expected to remain available, and incident management is the process that restores those services when they fail. Chen et al. describe it as detecting a service problem, creating an incident, determining its cause, restoring the service, and capturing knowledge to prevent a recurrence [1]. Incident triage is the stage at which a reported incident is assessed and directed towards resolution [1, 2]. It is a set of decisions made before the cause of the incident is known.

These decisions are difficult because the evidence available when they are made is often incomplete or conflicting. In a two-year study of incident management at Microsoft, Chen et al. found that distinct problems can produce similar symptoms, so that the symptoms alone may not identify the responsible team, and incidents are reassigned several times as a result [1]. Remil et al. note that data in real IT environments is commonly noisy and incomplete [2]. At the Directorate of University ICT Services (UIS) at Uganda Christian University, incidents reach developers through user tickets that are vetted and assigned, and application errors are diagnosed from system logs and a monitoring dashboard. An initial interview with a senior developer of the Directorate in October 2026 described this process and identified the application programming interface logs as the most informative source of evidence.

Reinforcement learning (RL) offers a way to treat triage as a sequence of decisions. In RL, an agent learns a policy, which is a rule for choosing actions, by acting in an environment and receiving rewards that reflect the consequences of its actions [3]. When the agent cannot observe the true state, it can choose actions for the information they yield, and weigh the cost of that information against its value [4]. Under this framing, an agent faced with an ambiguous incident can gather more evidence, recommend a fix, or hand the incident to a person.

1.2 Problem Statement

IT support teams must decide how to handle an incident before they know its cause. A member of IT staff facing such an incident can investigate further, apply a likely remedy, or pass the incident to someone better placed to resolve it, and each choice has a cost. Further investigation takes time, a wrong remedy can prolong or worsen the incident, and unnecessary escalation consumes the attention of senior staff.

Existing automated approaches address parts of this decision but not the choice among its options. Triage models commonly produce one prediction per incident report and are evaluated

with classification metrics [1,2], so they cannot ask for more evidence and cannot say that they are unsure. Reinforcement learning has been applied to incident response and investigation, mostly in cybersecurity [5–7], and a few studies of security alert triage let an agent gather evidence and then escalate or defer an alert to an analyst [8,9]. Their agents close or escalate alerts, however, and do not recommend a fix that changes the affected system. A systematic review of 34 studies, conducted for this research, found escalation to a person in only seven of them, and only three that report what happens when a rule mechanism blocks a decision [10]. A reward penalty alone does not ensure that an agent respects an organisation's operational rules, because penalising an action does not guarantee that the agent never selects it [11].

As a result, there is no evaluated agent for IT service incidents that learns when to investigate, when to recommend a permitted fix and when to escalate, with its proposals checked against explicit operational rules, and no evidence on how such an agent behaves as the evidence becomes more ambiguous. This research seeks to address that gap. It does not claim that constraining a learned incident-handling agent is new; its contribution lies in the decision it studies, the setting, and the way the result is evaluated.

1.3 Research Objectives

The general objective of this study is to design and evaluate a rule-constrained reinforcement learning agent for ambiguity-aware IT incident triage that learns when to gather additional diagnostic information, recommend permitted actions, or escalate incidents to a human expert. It is guided by the following specific objectives.

1.3.1 Scenario Model and Simulated Environment:

To build, by 1 November 2026, a simulated triage environment from a scenario model of about 15 causes, 20 diagnostic checks and 10 operational rules elicited from the Directorate and reviewed by its staff, in which an agent can request diagnostic information, recommend permitted actions or escalate incidents, and in which missing, unreliable and overlapping evidence can each be set independently. This objective addresses how ambiguous triage scenarios can be represented in a controlled environment.

1.3.2 Rule-Constrained Agent:

To train, by 8 November 2026, a deep Q-network agent over ten random seeds that chooses among requesting checks, recommending fixes and escalating, under an independent rule-based safety mechanism that returns a blocked proposal to the agent once and escalates the incident if the second proposal is also blocked, so that no prohibited action is executed. This objective addresses how reinforcement learning can support the three decisions and how effectively a rule mechanism prevents prohibited actions.

1.3.3 Evaluation against Baselines and under Ambiguity:

To evaluate the agent, by 22 November 2026, against six baseline decision policies on decision quality, investigation efficiency, escalation behaviour and constraint compliance, reported with 95% confidence intervals, and to measure how its accuracy and its mix of investigating, recommending and escalating change across controlled levels of missing, unreliable and overlapping evidence. This objective addresses how the approach compares with simpler policies and how robust it is as the evidence becomes more ambiguous.

1.4 Justification

This research is justified by a decision that IT support teams face routinely and that existing automated approaches represent only in part. The literature shows that incident evidence is often insufficient to identify the cause [1], and that automation in AIOps mainly stops at triage or the recommendation of solutions, with automatic execution limited and unverified [12]. Reinforcement learning and explicit rules compensate for each other's weaknesses: the learner can judge when further investigation is worth its cost, which fixed rules handle poorly, and the rules give an assurance about prohibited actions that a learned policy cannot [13]. Keeping the rules in a mechanism separate from the learner also makes them readable and changeable by IT staff.

The study is feasible and low-risk because it is carried out entirely in simulation. Simulation allows the level of ambiguity to be controlled and experiments to be repeated, without acting on any live system of the university. Working with the Directorate grounds the simulated scenarios and rules in the practice of an operating IT support unit. The study also supports the use of artificial intelligence in Uganda's public and institutional IT services, which Appendix F relates to the programme specialisation and to national and global development goals.

Chapter 2

Literature Review

2.1 Literature Review Plan

The review was planned around the decision this research studies: what an agent should do with an IT incident whose cause is uncertain. It drew on two sources. The first was a core set of publications on AI for IT operations (AIOps), safe reinforcement learning and learning to defer, collected while the problem was being defined. The second was a systematic review that followed the PRISMA 2020 guideline [10, 14]. The systematic review searched Scopus, the ACM Digital Library and IEEE Xplore on 23 October 2026 with the strings in Appendix E. It identified 4,399 records, screened 131 on title and abstract, assessed 40 reports in full, and included 34 studies that apply reinforcement learning or sequential decision-making to IT or security incident handling.

The literature is discussed chronologically, in the order in which the strands developed: the foundations of safe reinforcement learning, then information gathering and deferral, then the prediction-based approaches of AIOps, then reinforcement learning for incident handling, and finally the recent studies that begin to combine these strands. Each approach is examined against the requirements of IT incident triage at a university IT directorate, and the chapter closes with the research gaps.

2.2 Review of Approaches to Incident Triage

2.2.1 Foundations of Safe Reinforcement Learning (2015–2018)

These methods keep a learning agent from taking prohibited actions.

- **Constrained optimisation (2015–2017):** García and Fernández divided safe reinforcement learning into methods that change the optimisation criterion and methods that change exploration with external knowledge [15]. Constrained policy optimisation bounds the expected cumulative cost of a policy [16]. An expected-cost bound does not, however, enforce a rule that must hold at every step [17], and a reward penalty discourages an action without preventing it [11].
- **Shielding (2018 onwards):** Alshiekh et al. introduced a shield, a separate component that checks each action the agent proposes and blocks or replaces it when it would violate a specification [13]. Later work found that a shield which silently replaces actions can hinder learning, because the agent does not know it was overridden [18]. Shields were demonstrated mainly in control tasks and games [11,13], not on the rules of IT operations.

2.2.2 Information Gathering and Deferral (2018–2023)

These methods let a model delay or hand over its decision.

- **Learning to defer (2018):** Madras et al. trained a model to hand a case to a human decision-maker when the person is likely to do better [19], and Joshi et al. later extended deferral to sequential settings [20]. Selective prediction gives a model the related option to abstain when its confidence is low [21]. In each case the choice is between deciding and handing over; gathering more evidence first is not an option.
- **Information gathering (2019):** Janisch et al. trained an agent to acquire costly features one at a time before it classifies [22], building on the principle that, under partial observability, an action may be chosen for the information it yields [4]. Here the choice is between looking further and deciding; handing over is not an option.

2.2.3 Triage as Prediction in AIOps (2020–2025)

These methods treat triage as the prediction of a label from the incident report.

- **Incident management at scale (2020):** Chen et al. studied two years of incident management at Microsoft. They found that distinct problems can produce similar symptoms, so that incidents are reassigned several times, and described a recurrent model that predicts the responsible team from the incident text [1].
- **AIOps surveys (2024–2025):** Remil et al. catalogued the methods for classifying, prioritising and assigning incidents, and criticised the reliance on classification metrics that ignore the cost of a wrong action [2]. Zhang et al. observed that language-model assistants extended this work to diagnosis and mitigation suggestions, but that automation mainly stops at triage or recommendation [12]. These models return an answer for every report; they cannot state that more evidence is needed or that a person should decide.

2.2.4 Reinforcement Learning for Incident Handling (2023–2026)

These studies treat incident handling as a sequence of decisions.

- **Root cause analysis (2023):** Wang et al. used hierarchical reinforcement learning to decide which questions to put to engineers when building a dependency graph for root cause analysis [23], and Ding et al. learned to prune that graph before analysis [24]. Both treat investigation as a decision with a cost.
- **Security incident response (2025–2026):** Dunsin et al. trained an agent to conduct a malware investigation step by step [5]. Hammar and Li planned incident responses while holding a belief over the model of the incident itself [6], and Hammar et al. adapted response policies to system change within seconds [7]. Their agents choose among responses or investigative steps, but none can hand the incident to a person.

2.2.5 Recent Combinations (2025–2026)

The most recent studies begin to combine the strands above, mostly in security operations.

- **Gathering evidence and escalating in alert triage:** Okafor trained agents that dismiss, investigate or contain security alerts, with escalation to analysts fixed by design [25]. Sahin and Mert trained an agent that queries logs and threat intelligence before it escalates or closes an alert, with a separate layer that defers uncertain alerts to an analyst [8],

and Kovtun et al. learned a policy that chooses among shallow triage, deep analysis and deferral under analyst-capacity constraints [9]. Their final actions close, prioritise or escalate an alert, however, and none recommends a fix that changes the affected system.

- **Rule mechanisms in incident response:** Ravi et al.'s shield for industrial intrusion response replaced blocked actions and produced many more false actions than an unshielded agent [26]. Singh's deterministic guard for cloud remediation hands an action to a human when it fails, and raised governance compliance from 72.3% to 99.9% [27]. Alauthman et al. proposed a shielded agent for incident response in industrial control systems, but reported an evaluation plan and no measured values [28]. Of the 34 studies in the systematic review, only three say what happens to a blocked decision, and none lets the agent propose again [10].

2.2.6 Evaluation and Robustness

These studies show how such agents have been evaluated, and where the evaluations fall short.

- **Measures:** In the systematic review, 31 of 34 studies measure decision quality, but only five measure escalation behaviour and four constraint compliance; one study, Kovtun et al., reports all four kinds of measure, for SOC alert escalation [9, 10]. A single metric can hide a degenerate policy: Okafor found that F1 and ROC-AUC concealed agents that had collapsed into trivial detectors, which the Matthews correlation coefficient exposed [25].
- **Robustness:** Fifteen of the reviewed studies test their agents under changed conditions, and the methods that keep a belief over the hidden state degrade least [6, 7]. Sahin and Mert show that deferral to analysts rises as the evidence degrades [8]. No study reports how the mix of investigating, acting and escalating changes as ambiguity increases.

Table 2.1 compares the studies closest to this research against the capabilities it requires.

Table 2.1: Comparison of the studies closest to this research. ✓ = addressed; ○ = partly addressed; – = not addressed.

Study	**Setting**	**Seq.**	**Info.**	**Rule**	**Esc.**	**Amb.**
Chen et al. [1]	IT incident triage	–	–	–	–	–
Wang et al. [23]	Root cause analysis	✓	✓	–	○	–
Dunsin et al. [5]	Malware investigation	✓	✓	–	–	–
Hammar and Li [6]	Security incident response	✓	–	–	–	✓
Alauthman et al. [28]	Industrial control incident response	✓	–	✓	○	–
Ravi et al. [26]	Industrial control incident response	✓	–	✓	–	–
Okafor [25]	Security alert triage	✓	✓	○	○	–
Sahin and Mert [8]	Security alert triage	✓	✓	○	✓	✓
Kovtun et al. [9]	Security alert escalation	✓	○	○	✓	–
Singh [27]	Cloud operations remediation	○	–	✓	○	–
This study	IT incident triage	✓	✓	✓	✓	✓

Seq. = sequential policy; Info. = information gathering; Rule = rule check; Esc. = human escalation; Amb. = varied ambiguity.

2.3 Research Gaps

The approaches reviewed above each solve part of the triage decision, but no study combines them for IT service incidents. The most significant gap is in the decision itself: information-gathering studies let an agent choose between looking further and deciding, deferral studies let it choose between deciding and handing over, and the security alert-triage studies that join the two do not recommend a fix that acts on the system. No reviewed study trains one policy to choose among investigating, recommending a remedy and escalating for IT service incidents. The second gap concerns the rules: hard rule mechanisms have been measured in only two studies, their results point in opposite directions, and no study gives a blocked agent a second proposal or reports how often blocked proposals occur during training. The third gap concerns robustness: decision accuracy and the rate of deferral have been tested under degraded evidence, but not the mix of investigating, recommending and escalating as ambiguity rises. Underneath all three lies a gap in representation, since the available environments model cyberattacks and microservice faults, and none lets the ambiguity of IT service incidents be varied.

These gaps are stated for the literature reviewed. The systematic review searched three databases, limited its IEEE Xplore search to open-access records, did not carry out citation searching, and could not retrieve seven reports [10]. The claim is that the reviewed literature leaves these questions open, not that they have never been examined.

Chapter 3

Methodology

3.1 Overview

To study how an agent should handle IT incidents whose cause is uncertain, this research adopts a design-and-evaluate approach [29]. It builds an artefact, a rule-constrained reinforcement learning agent together with the simulated environment in which it acts, and evaluates it in controlled experiments. The work proceeds in eight phases, shown in Figure 3.1: specifying the problem, eliciting scenarios with the Directorate, building and validating the environment, encoding the rules, implementing the baselines, training the agents, running the experiments, and analysing the results.

The evaluation is conducted entirely in simulation, for two reasons. Realistic incident data is scarce and confidential [2], and simulation lets the level of ambiguity be set, held constant and varied, which live incidents would not allow. The agent is never connected to any system of the university. The framework will be implemented in Python, with the Gymnasium interface for the environment [30] and the Stable-Baselines3 library for the learning algorithms [31]. No personal data, credentials or confidential details of the university's systems will enter the simulator, and staff participation will be voluntary and with consent.

3.2 Data Collection

The data for this research is a scenario model elicited from the Directorate of University ICT Services, not a dataset of incident records. The model will be elicited from Directorate staff using a structured workbook, and it will contain about 15 causes in four provisional categories (application and interface errors, authentication and access failures, database failures, and failures of infrastructure and external services), each with one correct fix; 20 diagnostic checks, such as an entry in the API logs, a value on the monitoring dashboard or the result of reproducing a failing request, each with an effort cost; a fixed list of symptoms; and 10 operational rules.

For each pair of cause and check, staff will rate how often the check is positive when that cause is present, on a five-point scale (never, rarely, sometimes, often, almost always) that is mapped to the probabilities 0.05, 0.25, 0.5, 0.75 and 0.95. Asking for a verbal rating keeps the 300 judgements manageable for practitioners. Any parameter that the elicitation does not supply will be set by the researcher and reported as such. After the environment is built, sample incidents will be shown to Directorate staff, who will judge whether they are plausible and whether the rules are stated correctly. The model will be revised and then frozen before any

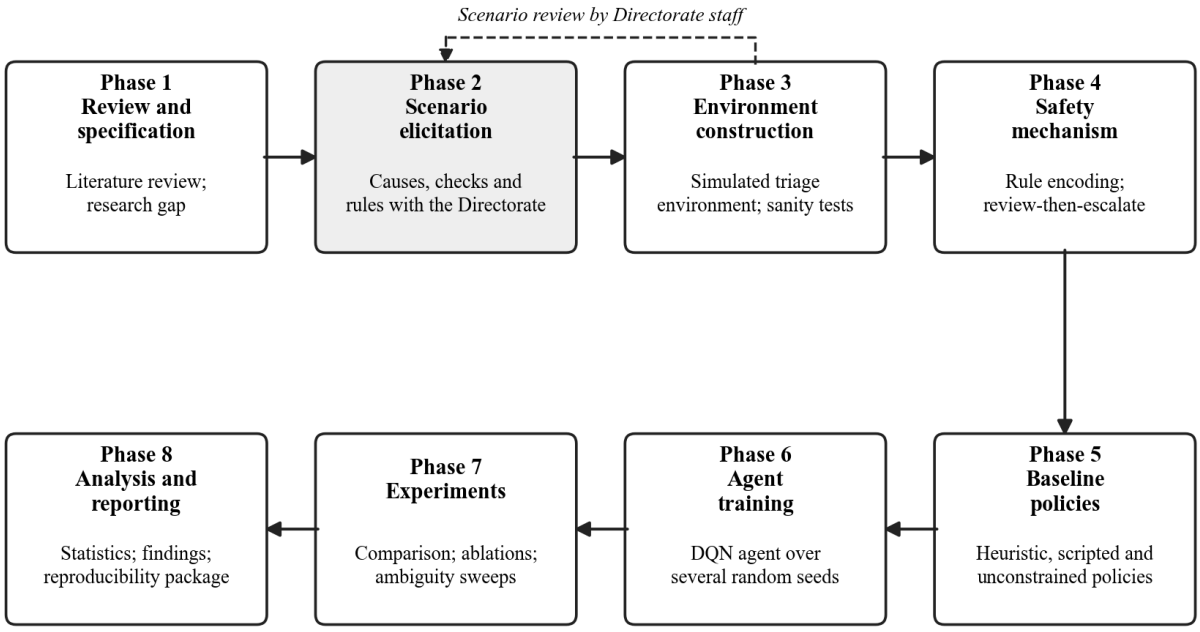


Figure 3.1: Research pipeline

agent is trained, so that the scenarios cannot be adjusted in favour of the proposed agent.

3.3 The Simulated Triage Environment

The environment casts triage as a sequential decision problem under partial observability [4]; one episode is the triage of one incident.

- **Hidden incident:** at the start of an episode a cause is drawn from the elicited prior and kept hidden from the agent. A visible context, such as whether the affected system is in production, is also drawn, and the rules refer to it. The result of every check is drawn once from the elicited likelihoods, so that requesting a check twice returns the same result.
- **Observation:** the agent sees a one-hot vector recording, for each check, whether it is not yet requested, unavailable, negative or positive, together with the symptoms, the context, the number of steps taken, and a flag set when its previous proposal was blocked. Because the checks are fixed within an episode and independent given the cause, this vector is a sufficient statistic for the belief that a partially observable agent would maintain.
- **Actions:** 36 actions: 20 that each request one check, 15 that each recommend the fix for one cause, and one that escalates the incident. A recommendation or an escalation ends the episode, and reaching the step limit counts as an escalation.
- **Reward:** a correct fix earns $+r_{\text{correct}}$, a wrong fix costs λ_{wrong}, each check costs k_i, and an escalation costs λ_{esc}, with $r_{\text{correct}} > -\lambda_{\text{esc}} > -\lambda_{\text{wrong}}$, so that resolving correctly is better than escalating and escalating is better than a wrong fix. The starting values are $+10$, -3 and -10, with check costs between 0.1 and 1.0, and they will be tuned in pilot runs.
- **Ambiguity controls:** missing evidence ($p_{\text{miss}} \in \{0, 0.2, 0.4, 0.6\}$) makes a check unavail-

able; unreliable evidence ($p_{\text{noise}} \in \{0, 0.1, 0.2, 0.3\}$) reverses its result; and overlapping evidence ($\omega \in \{0, 0.25, 0.5, 0.75\}$) moves each cause's likelihoods towards the average of its category, $\theta'_{ij} = (1 - \omega)\theta_{ij} + \omega\bar{\theta}_i$, so that causes become harder to tell apart.

- **Reference labels:** for evaluation only, each incident is labelled as one that should be escalated if the rules forbid the fix for its true cause, or if it stays ambiguous even with every check observed. Before training, the simulator will be checked for leakage of the hidden cause into the observations, a defect that Sahin and Mert found in an earlier version of their own simulator [8].

3.4 The Reinforcement Learning Agent

3.4.1 Deep Q-Network:

The primary learner is the deep Q-network (DQN), which estimates the expected return of each action with a neural network [32]. It suits the task because the action set is small and discrete, the 4^{20} combinations of check results rule out a lookup table, and the value it assigns to each action, such as a blocked fix against escalation, can be inspected. The network has two hidden layers of 128 units. Actions that are not available, such as checks already requested and blocked proposals, are masked both when an action is chosen and in the maximum of the learning target.

3.4.2 Proximal Policy Optimisation:

Proximal policy optimisation (PPO) with action masking will serve as a secondary learner, to show whether the findings depend on the choice of algorithm [33]. Each agent will be trained with ten random seeds, at a fixed base level of ambiguity and, in a second regime, with the ambiguity controls drawn at random for each episode.

3.5 The Rule-Based Safety Mechanism

The safety mechanism is a component separate from the agent that checks every proposed action against the ten operational rules. Each rule is a readable condition over the proposed action, the context of the incident and the evidence gathered so far; for example, a rule may forbid a fix on a production system, or forbid a fix until a named check has been observed. Escalation is never forbidden, so a permitted action always exists. When the agent proposes an action, a permitted action passes to the environment unchanged. A prohibited action is blocked and logged, and the agent sees the blocked flag, with that action masked, and proposes once more. If the second proposal is also prohibited, the incident is escalated to the human expert and recorded as a forced escalation. Telling the agent about the block responds to the finding that a shield which silently overrides actions can hinder learning [18]. No study in the systematic review gives a blocked agent a second proposal [10].

Figure 3.2 shows how the agent, the safety mechanism and the simulated environment work together.

3.6 Baseline Decision Policies

The agent will be compared with six baseline policies that run in the same environment on the same evaluation incidents, each standing for an approach from Chapter 2 (Table 3.1). Two

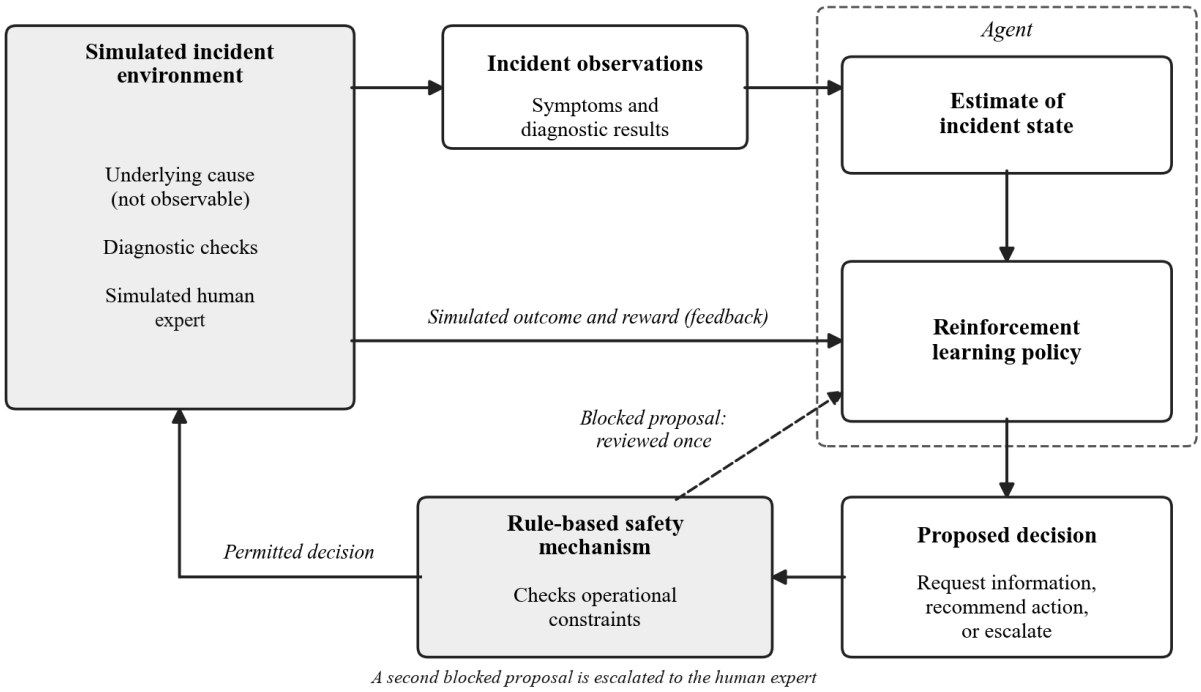


Figure 3.2: Proposed Architecture

ablations of the agent, one that cannot request checks and one that cannot escalate, will show what each option contributes.

3.7 Evaluation Strategy

The framework will be evaluated across four groups of measures. Decision quality will be measured by resolution accuracy, coverage, selective accuracy and the wrong-fix rate. Investigation efficiency will be measured by the mean number and cost of checks and an accuracy–cost curve [22]. Escalation behaviour will be measured by the escalation rate and by the precision, recall, F1 score and Matthews correlation coefficient of escalations against the reference labels [25]. Constraint compliance will be measured by the proposed-violation rate, the forced-escalation rate, the executed-violation rate and the compliant success rate [34].

Four experiments use these measures. Experiment 1 compares the agent with the six baselines at the base level of ambiguity. Experiment 2 compares it with the unconstrained and penalty-only agents and tracks the proposed-violation rate over training. Experiment 3 compares it with its two ablations. Experiment 4 evaluates every policy across the planned ambiguity levels, for both training regimes. A sensitivity analysis varies the cost of escalation. Results will be reported as the mean over ten seeds with 95% bootstrap confidence intervals [35], on evaluation incidents generated with seeds never used in training.

Table 3.1: Baseline decision policies

Policy	**Behaviour**	**Approach represented**
Initial-symptom	Recommends the most likely cause from the first symptoms, without checks	Triage as one prediction
Scripted checklist	Requests a fixed sequence of checks for the category, elicited from Directorate practice, then recommends	Runbook practice
Always-escalate	Escalates every incident	The limit of deferral
Confidence threshold	Tracks the posterior over causes, requests the most informative check until one cause passes a threshold, else escalates	Uncertainty-driven investigation (knows the model)
Unconstrained agent	The same DQN without the safety mechanism	Standard shielding comparison
Penalty-only agent	The same DQN without the mechanism, with a reward penalty per rule violation	Constraint by reward penalty

Chapter 4

Expected Results

4.1 Decision Quality

The agent is expected to achieve higher decision quality than the selected baseline policies in ambiguous incident scenarios (hypothesis H1). Specifically, it is expected to resolve a larger share of incidents correctly than the initial-symptom, scripted and always-escalate policies at equal or lower investigation cost, and to approach the confidence-threshold policy without being given the likelihood model. Numerical targets will be fixed after the pilot runs, and the results will be reported whether or not the hypothesis holds.

4.2 Rule Compliance

With the safety mechanism in place, the agent is expected to achieve a higher rate of rule-compliant correct recommendations than the same agent without it (hypothesis H2). No prohibited action will be executed when the mechanism is active, so the comparison of interest is the compliant success rate against the unconstrained and penalty-only agents. The proposed-violation rate is expected to fall over training, which would show that the agent learns to act within the rules and does not rely on being stopped.

4.3 Robustness to Ambiguity

As the level of ambiguity rises, the agent is expected to request more diagnostic information and to escalate more incidents, and not to continue recommending fixes on weaker evidence (hypothesis H3). The agent trained across randomised ambiguity levels is expected to degrade less than the agent trained at a single level. The experiments will report what no reviewed study has reported: how the shares of investigating, recommending and escalating change across controlled levels of missing, unreliable and overlapping evidence.

4.4 Prototype

The research is expected to produce a working simulator of IT incident triage built from the Directorate's scenarios, the rule-based safety mechanism, six baseline policies, trained agents, and scripts that reproduce every experiment. A lightweight dashboard will replay an incident step by step, showing the evidence revealed, the agent's estimated action values, any blocked proposal and the final decision, beside the comparison results.

Broader impact. Putting this together, the expected results address the problem this research is concerned with: that automated triage tools predict an answer for every incident and do not know when to look further, when to act within the rules, and when to hand the incident to a person. An agent that makes these three decisions under explicit, readable rules, evaluated on all four groups of measures, would give IT support teams such as the Directorate evidence on how far such automation can be trusted before it is considered for use.

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Appendix A

Budget

The research has few direct costs. Training and evaluation will run on the university's computing facilities and the researcher's laptop with open-source software (Python, Gymnasium, Stable-Baselines3 and PyTorch), and the elicitation sessions with the Directorate of University ICT Services involve no cost. The only expected cost is printing. Unit costs follow the author's earlier proposal budget and are estimates to be confirmed.

No.	**Item**	**Details**	**Unit Cost (UGX)**	**Qty**	**Total (UGX)**
1	Project report printing and binding	3 copies for the supervisor, department, personal	50,000	3	150,000
2	Proposal printing	2 copies for supervisor review	10,000	2	20,000
				TOTAL BUDGET (UGX)	**170,000**

Appendix B

Research Project timelines

The project is managed in seven phases. The implementation runs over six weeks, from 19 October to 29 November 2026, after the literature review and the stakeholder interview. Each phase depends on the one before it, because the environment cannot be built before the scenarios are specified, and the agents cannot be trained before the rules and baselines exist.

Project Timeline (19 October to 29 November 2026)

Phase	Phase Name	Key Activities	Weeks	Depends on	Deliverable
1	Literature and Planning	Thematic literature review and Chapters 1–3; stakeholder interview with the Directorate (19 Oct); systematic literature review (23 Oct); approvals; elicitation workbook.	Before 19 Oct, Week 1	–	Proposal, systematic review, elicitation workbook.
2	Scenario Elicitation	Elicitation session(s) with Directorate staff; scenario model of about 15 causes, 20 checks and 10 rules written up as configuration files.	Weeks 1–2	1	Scenario model v1.
3	Environment and Rules	Incident generator and Gymnasium environment with the three ambiguity controls; the rules and the review-then-escalate mechanism; tests; staff review of sample incidents; model frozen.	Weeks 1–2	2	Frozen scenario model; tested simulator and rule mechanism.
4	Baselines and Training	Six baseline policies and the metrics module; pilot runs; small hyperparameter search; the agent, unconstrained and penalty-only agents trained over ten seeds.	Weeks 2–3	3	Baselines run end to end; trained agents.

Continued on next page

Phase	Phase Name	Key Activities	Weeks	Depends on	Deliverable
5	Experiments	Experiments 1–3 (baselines, effect of the mechanism, ablations); sensitivity to escalation cost; Experiment 4 (robustness across ambiguity levels).	Weeks 4–5	4	Results tables and charts.
6	Analysis and Prototype	Confidence intervals, decision-mix charts and findings against the hypotheses; replay dashboard and results panel.	Weeks 5–6	5	Findings summary; working prototype.
7	Writing and Submission	Results chapter, revision of Chapters 1–3, poster, archive of code and configurations, presentation rehearsal.	Week 6	6	Poster, report, archived code.

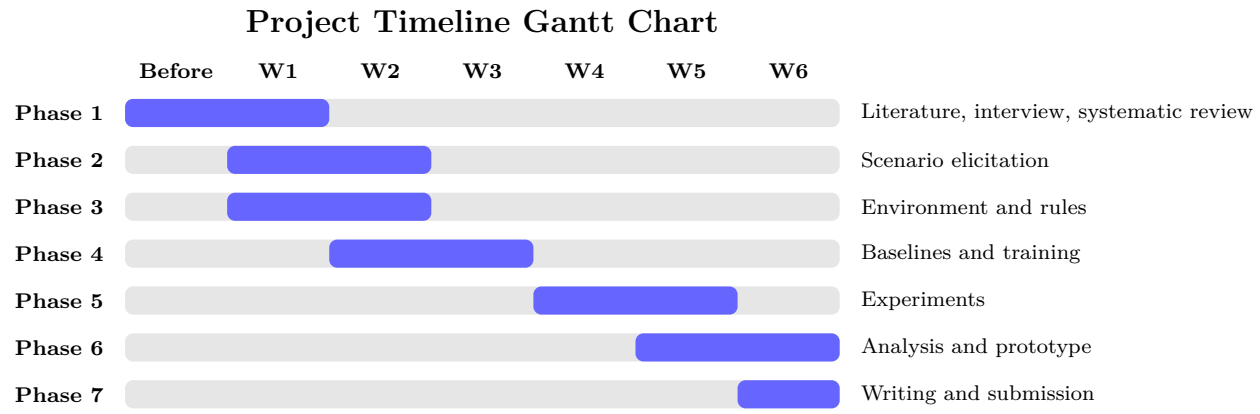


Figure B.1: Research project timeline Gantt chart. W1 begins on 19 October 2026 and W6 ends on 29 November 2026.

Appendix C

Declaration of AI use

Declaration Statement:

I, *OPOL CHARIS*, declare that I used

i. ***Claude by Anthropic (Claude Opus 5.5, through Claude Code)*** to ***design and carry out the systematic literature review***: drafting the title, keywords, research questions, search strings and eligibility criteria with me; writing the scripts that merged the database exports, removed duplicates and applied the automated filters; making the title and abstract screening decisions; assessing the full texts I retrieved, extracting their data and rating their quality; and drafting the text of the review. I ran the database searches, retrieved the full texts, recorded which reports were closed access, and made the full-text decisions on two reports. Prompts used: *"I need to work on a PRISMA guided systematic literature review for this project. I need you to help start by giving me the SLR title, keywords, objectives, research questions and how they come back to the objectives and the search string, databases to search and an exclusion and inclusion strategy"*; *"I have exported the results, could you go through them the PRISMA process, when you are done give me the list of papers that I should download"*; *"Use this sample SLR and write the introduction and methodology … Now don't hallucinate."*

ii. ***Claude by Anthropic*** to ***check my thesis chapters against the systematic review*** and to propose and apply revisions to Chapters 1–3, including the comparison of the closest studies and the statement of the research gaps. Prompts used: *"From everything we have discussed, what could you update in the thesis to make it stronger"*; *"Okay go ahead and do that."*

iii. ***Claude by Anthropic*** to ***draft this proposal*** by condensing my thesis Chapters 1–3 and the implementation concept note, and to lay it out on the department template. Prompts used: *"Work on a proposal like the one I shared"*; *"Use the exact same document formatting because it was a template."*

iv. ***Claude by Anthropic*** to ***prepare a reading plan*** of the papers I should read in full, and to ***adapt my earlier research poster*** to this project. Prompts used: *"What are the most important papers I should read to ground myself in the entire work"*; *"Create a reading plan MD with checklist with all of this"*; *"I want you to edit [the poster] to this project."*

Add any other AI use in earlier drafts of the thesis chapters, and state how you modified the outputs.

Signature: _____ Date: _____

Appendix D

List of 10 Major Journal Publications reviewed

The publications are listed in order of their relevance to this research: three review papers, four journal articles and three conference papers. Fill in the impact factors from the Journal Citation Reports or Scopus; they have not been verified.

Publications Ranked by Relevance

Rank	Title	Journal / Venue	Type	Impact Factor
1	Sahin & Mert (2026): Dual-Threshold Conformal Deferral for Trustworthy Security Alert Triage [8]	Electronics	Journal Article	TBC
2	Kovtun et al. (2026): Risk-Sensitive Decision-Making in SOC Alert Escalation under Resource Constraints [9]	CEUR Workshop Proceedings	Conference Paper	TBC
3	Okafor (2026): A Leakage-Free Benchmark of Offline Policy Learning for SOC Alert Triage [25]	Machine Learning with Applications	Journal Article	TBC
4	Hammar et al. (2026): Adaptive Network Security Policies via Belief Aggregation and Rollout [7]	IEEE Transactions on Information Forensics and Security	Journal Article	TBC
5	Chen et al. (2020): Towards Intelligent Incident Management [1]	ESEC/FSE	Conference Paper	TBC
6	Remil et al. (2024): AIOps Solutions for Incident Management [2]	arXiv	Review	TBC
7	Alshiekh et al. (2018): Safe Reinforcement Learning via Shielding [13]	AAAI Conference on Artificial Intelligence	Conference Paper	TBC

Continued on next page

Rank	Title	Journal / Venue	Type	Impact Factor
8	Könighofer et al. (2025): Shields for Safe Reinforcement Learning [11]	Communications of the ACM	Review	TBC
9	García & Fernández (2015): A Comprehensive Survey on Safe Reinforcement Learning [15]	Journal of Machine Learning Research	Review	TBC
10	Mnih et al. (2015): Human-Level Control through Deep Reinforcement Learning [32]	Nature	Journal Article	TBC

Relevance Key

	Addresses reinforcement learning or sequential decisions in IT or security incident handling
	Provides the methods the study builds on: safe reinforcement learning, shielding, deep Q-networks

Appendix E

Scopus Search Strings used during the literature review

The systematic review searched three databases on 23 October 2026. The strings are given exactly as they were run.

E.1 Scopus and ACM Digital Library (all fields)

```
( "reinforcement learning" OR "deep Q-network*" OR "Q-learning" OR "policy gradient"
OR "actor-critic" OR "proximal policy optimi*" OR "Markov decision process*" OR
"partially observable Markov" OR POMDP OR "contextual bandit*" OR "multi-armed
bandit*" ) AND ( "incident management" OR "incident response" OR "incident triage"
OR "incident handling" OR "incident resolution" OR "incident mitigation" OR AIOps OR
"IT operations" OR "IT service management" OR ITSM OR "site reliability" OR "service
desk" OR "help desk" OR "trouble ticket*" OR "ticket routing" OR "ticket assignment"
OR "alert triage" OR "alert prioriti*" OR "security operations cent*" OR "automated
remediation" OR ( "root cause" AND ( microservice* OR cloud OR "online service*" OR
"IT system*" ) ) ) AND ( triage OR escalat* OR defer* OR "human-in-the-loop" OR
"human in the loop" OR "human oversight" OR investigat* OR diagnos* OR "information
gathering" OR "information acquisition" OR remediat* OR mitigat* OR constrain* OR
shield* OR "safe reinforcement learning" OR "action mask*" OR "partial* observ*" OR
uncertain* OR ambigu* ) AND ( "2015" OR "2016" OR "2017" OR "2018" OR "2019" OR
"2020" OR "2021" OR "2022" OR "2023" OR "2024" OR "2025" OR "2026" )
```

Records exported: Scopus 3,394; ACM Digital Library 1,000.

E.2 IEEE Xplore (open access only)

```
( "reinforcement learning" OR "deep Q-network*" OR "Q-learning" OR "policy gradient"
OR "actor-critic" OR "proximal policy optimization" OR "Markov decision process*" OR
"partially observable Markov" OR POMDP OR "contextual bandit*" OR "multi-armed
bandit*" ) AND ( "incident management" OR "incident response" OR "incident triage"
OR "incident handling" OR "incident resolution" OR "incident mitigation" OR AIOps OR
"IT operations" OR "IT service management" OR ITSM OR "site reliability" OR "service
desk" OR "help desk" OR "trouble ticket*" OR "ticket routing" OR "ticket assignment"
OR "alert triage" OR "alert prioritization" OR "security operations center" OR
"automated remediation" OR ("root cause" W/5 (microservices OR cloud OR "online
services" OR "IT systems")) ) AND ( triage OR escalation OR defer* OR
```

```
"human-in-the-loop" OR "human in the loop" OR "human oversight" OR investigation OR
diagnosis OR "information gathering" OR "information acquisition" OR remediation OR
mitigation OR constraint* OR shield* OR "safe reinforcement learning" OR "action
masking" OR "partial observability" OR uncertainty OR ambiguity )
```

Records exported: 5.

Appendix F

Research alignment with specialization, SDG, NDPIV & Vision 2040

F.1 Alignment with programme specialization

This research aligns with the Artificial Intelligence and Robotics specialization track of the Bachelor of Science in Computer Science. Reinforcement learning, in which an agent learns to act by interacting with an environment, is a core method of both robotics and artificial intelligence. The project applies deep reinforcement learning, decision-making under partial observability, and safe reinforcement learning, in which a learning agent is kept within explicit constraints, a problem first studied for robots and control systems [13].

F.2 Alignment with Sustainable Development Goals (SDG)

The research aligns with SDG 9, Industry, Innovation and Infrastructure, which calls for resilient infrastructure and inclusive innovation. Reliable IT services are part of an institution's infrastructure, and the project studies how automation can support the people who restore those services without acting outside the rules they set. It also supports SDG 4, Quality Education, since the university's teaching, learning and administration depend on the IT services that the Directorate maintains.

F.3 Alignment with the National Development Plan IV (NDPIV)

The National Development Plan IV continues Uganda's emphasis on the digital transformation of public and private services. A university IT directorate is one of the institutions through which this transformation is delivered, and the project examines a safe, rule-governed way of introducing artificial intelligence into its incident handling. Cite the specific NDP IV programme and objective that this supports.

F.4 Alignment with the Uganda Vision 2040

Uganda Vision 2040 aims to transform Uganda into a competitive upper-middle-income country and treats information and communication technology as one of the foundations of that transformation [36]. This research contributes locally developed expertise in artificial intelligence for IT operations, built with and for a Ugandan institution. Check the wording against the relevant section of Vision 2040 and add a page reference.

Appendix G

Algorithms, Code of interest

G.1 Masked Deep Q-Network Update

The agent is trained with the deep Q-network of Stable-Baselines3 [31]. Because that implementation does not support action masking, the target of each update is restricted to the actions available in the next state, $A(s')$, which excludes checks already requested and a proposal that was just blocked:

$$y = r + \gamma \max_{a' \in A(s')} Q_{\text{target}}(s', a').$$

The same restriction applies when the agent chooses an action. The PPO learner uses the MaskablePPO implementation of the sb3-contrib repository, which supports masking directly.

G.2 Review-then-Escalate Procedure

1. The agent proposes action a. If no rule prohibits a, it passes to the environment.
2. Otherwise the block is logged, the blocked flag is set, a is removed from the available actions, and the step is consumed without reward.
3. The agent proposes action a_2. If no rule prohibits a_2, it passes to the environment.
4. Otherwise the incident is escalated to the human expert and recorded as a forced escalation.

G.3 Code Repositories

- Stable-Baselines3 (DQN and PPO): `https://github.com/DLR-RM/stable-baselines3`
- sb3-contrib (MaskablePPO): `https://github.com/Stable-Baselines-Team/stable-baselines3-contrib`
- Gymnasium (environment interface): `https://github.com/Farama-Foundation/Gymnasium`

Appendix H

Other useful figures

Figure H.1 shows the flow of records through the systematic literature review that informed Chapter 2.

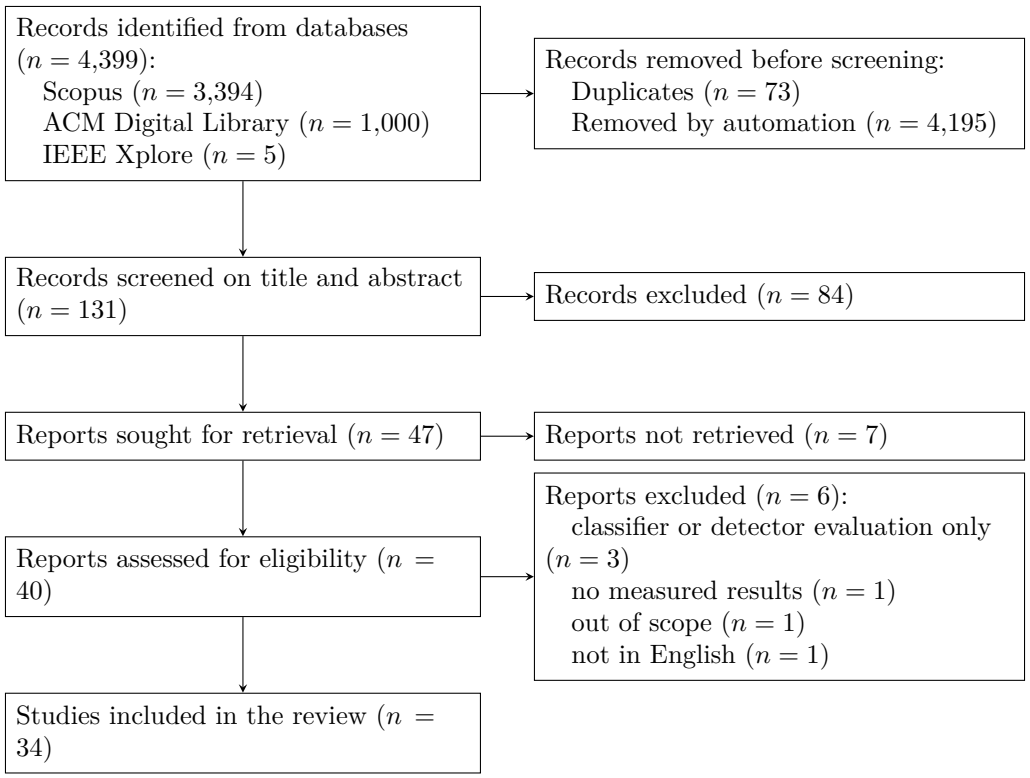


Figure H.1: PRISMA 2020 flow diagram of the systematic literature review. Citation searching was not carried out.

Appendix I

Research poster layout

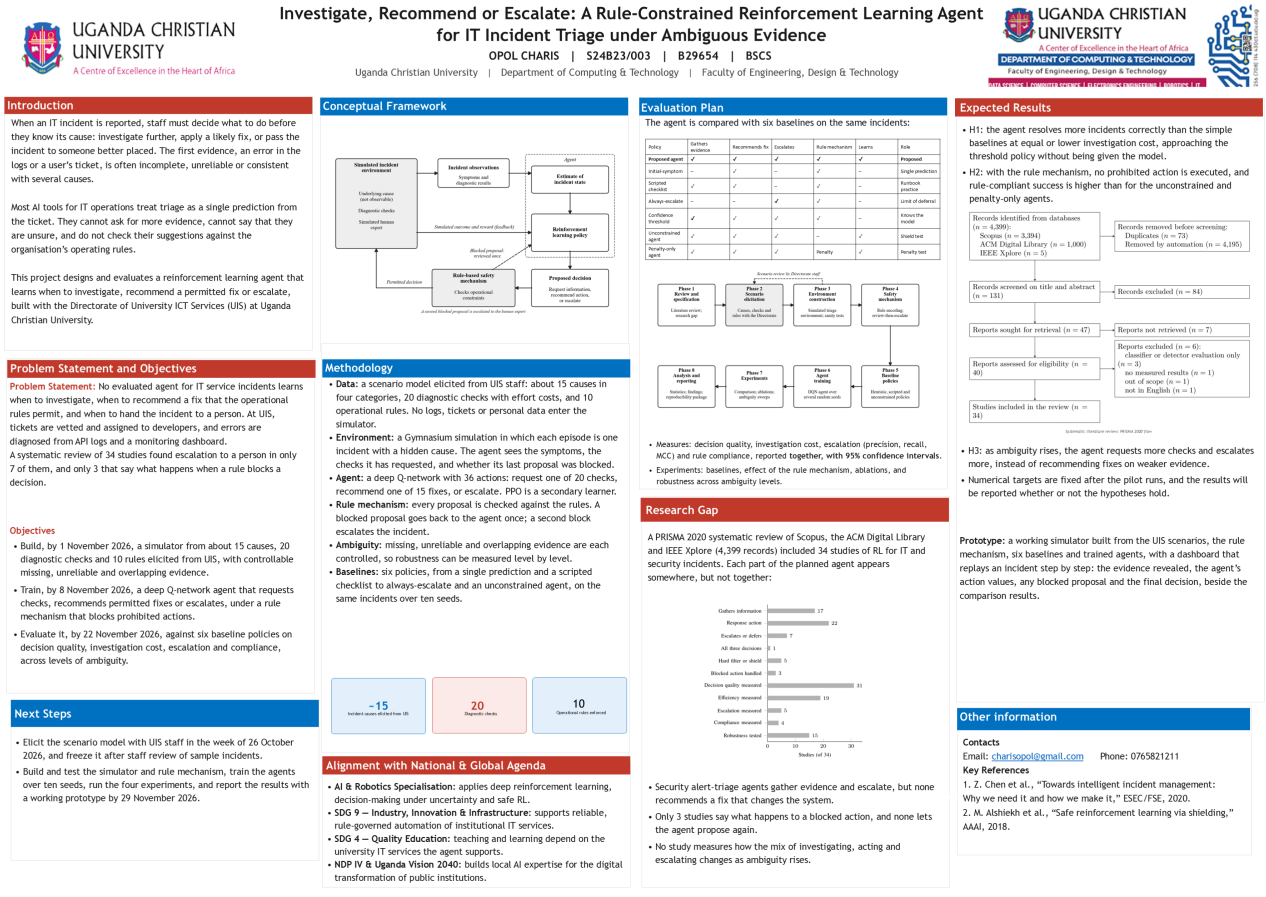


Figure I.1: Research proposal presentation poster